DISTRIBUTED REAL-TIME SYSTEMS

MotionTennis

M3 - System design and integration test specification

TEAM	MotionTennis
MILESTONE	M3
DATE	2026-11-10
REVISION	R4

Review status

Author consistency review completed. Team design approval and acceptance of the separate Mx - Integration Test Debugging Concept are pending. Gate 2 and M3 are not claimed as passed.

This is a self-contained, anonymous team report. Requirements, diagrams, budgets, interfaces, fault assumptions, test procedures and traceability are included. All hardware tests are planned, not executed.

1. System and current status

Applies: F-01, F-03, F-08, F-13, F-16-F-19; N-05, N-10.

Four people play 2v2 tennis using wireless, battery-powered ESP32 controllers. Each controller reads an MPU-6050 IMU over I²C using Zephyr. A webcam continuously observes body and arm movements. The first version uses calibrated 2.5D gestures; the Pi moves the virtual players automatically.

The **Raspberry Pi 3 B calculates** ball movement, player movement, hits, serves and score. A separate **C# service on the Windows laptop stores** the accepted match state. **Unity displays** that state in two team views. Tracking, storage, Unity and debugging are separate processes on the same laptop.

Current implementation evidence covers an in-memory C# state-store core and offline checks. An integrated network/game runtime, target compatibility and hardware timing are not established. [F-08, N-05, N-10]

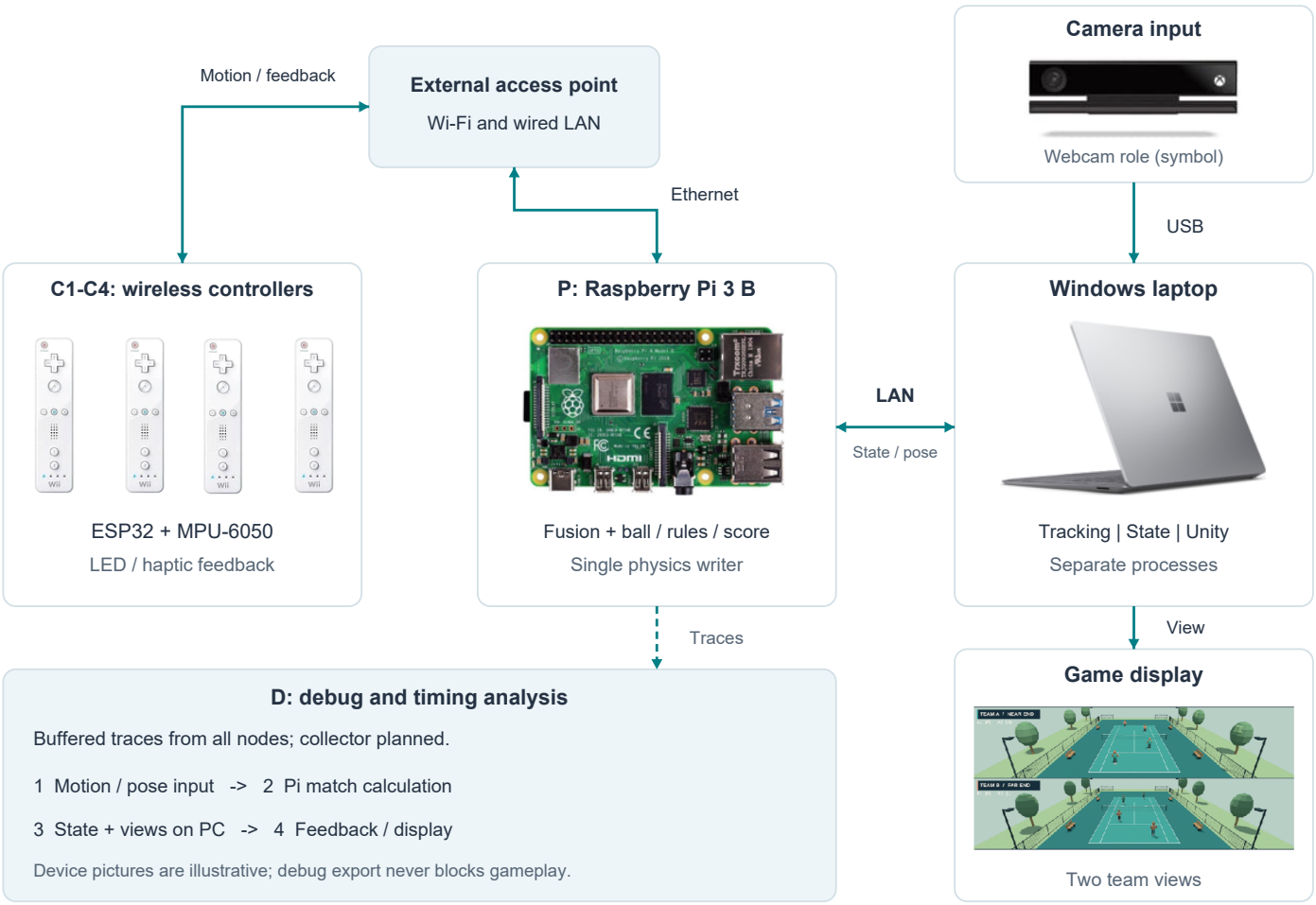
Status terms: agreed = recorded project decision; proposed = awaiting approval or measurement; open = unresolved. All timing numbers below are proposed targets, not measured guarantees.

2. Architecture

Diagram and deployment: F-01, F-03, F-08, F-12, F-13, F-16, F-19; N-05, N-06, N-09.

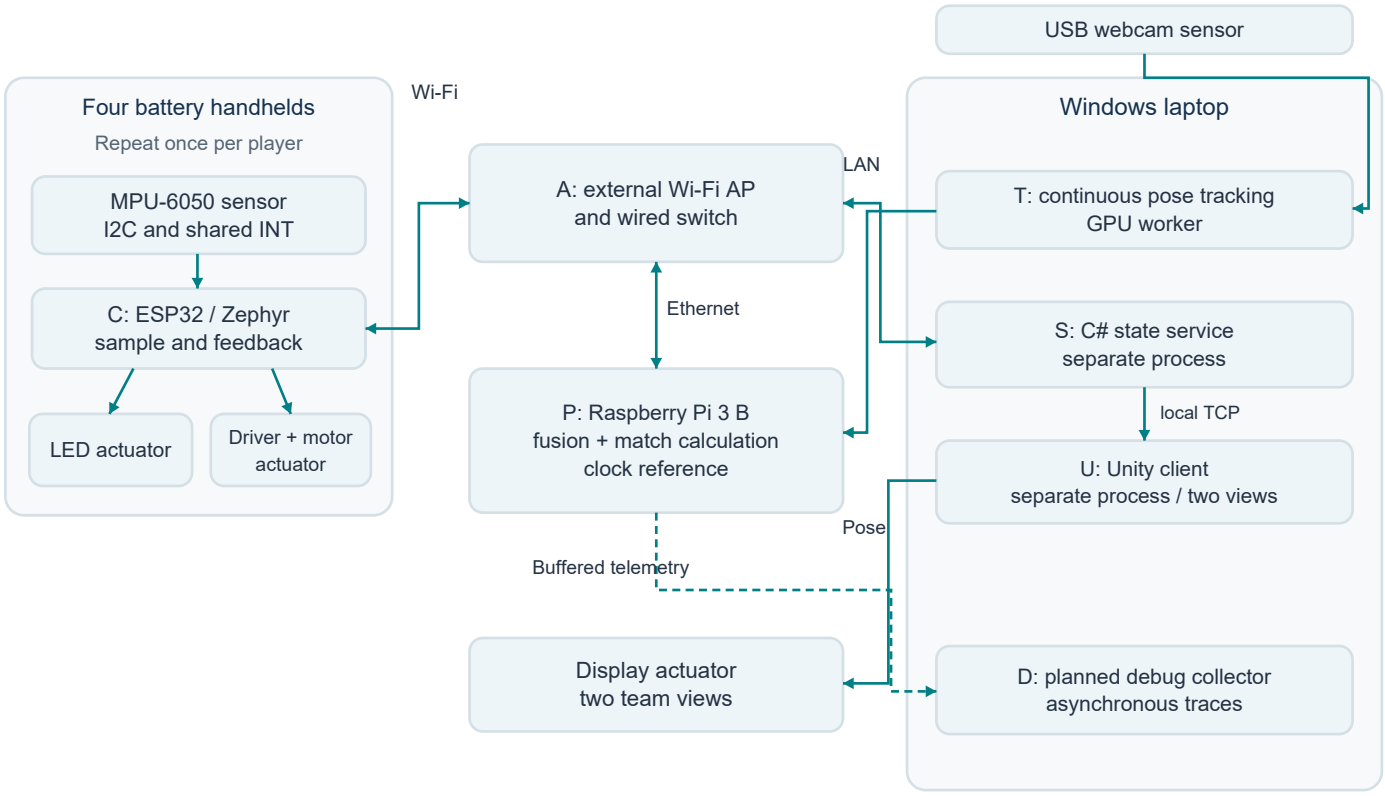
The diagram shows **variant A**: controllers use an external access point; Pi and laptop use wired Ethernet. **Variant B** lets the Pi provide the hotspot while the laptop stays wired. Compare both with identical messages, sample rates and tests. B must first prove that every controller can reach the Windows debug collector. No faster variant has been measured. [F-01, N-01, N-05]

Big picture



The supplied device pictures illustrate roles: the planned hardware remains ESP32/MPU-6050 controllers, a webcam and Pi 3 B. The game picture illustrates the two-team view; it is not runtime validation. The following technical view separates the software nodes and their links. [F-01, F-03, F-08, F-13, F-19; N-05]

Technical node view



All components export bounded traces to D; debug never blocks gameplay.

Variant B: Pi provides the Wi-Fi hotspot; the laptop remains wired.

Camera video stays on the laptop. The debug collector observes the system without blocking gameplay. The exact controller board, actuator, wiring, webcam and software versions must be pinned before hardware tests. [N-05, N-09]

3. Transactions and deadlines

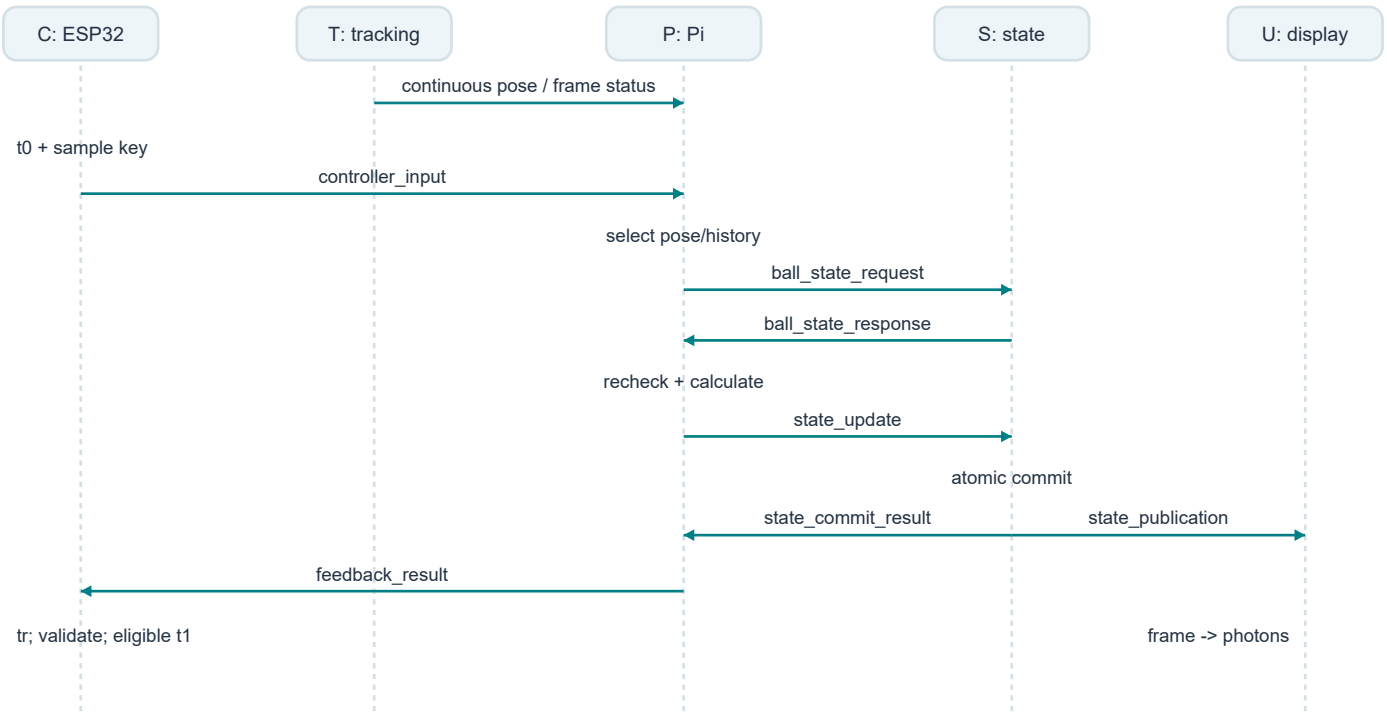
Applies: F-02-F-10; N-01-N-04, N-10; T-01-T-10; P-01, P-03.

A **transaction** is one sensor sample followed through the system. Its identity is **(controller ID, boot session, request sequence)**. Every regular sample gets its own identity, even when several samples share a network batch.

Transaction	Owner role	Start → end	Proposed deadline / outcome	Requirements
TX-01: controller feedback	Controller integration owner	Regular IMU boundary **t0** → valid matching response **tr** → eligible LED/motor electrical onset **t1**	Response p99 ≤30 ms; eligible electrical feedback p99 ≤40 ms. Local expiry is 50 ms: suppress new late gameplay cues	F-02-F-07; N-01; T-01, T-02
TX-02: visible event	Presentation integration owner	The same sample **t0** → accepted match event → photons in each team view	**P-01 proposal:** p99 ≤80 ms for selected visible events. This screen target is not yet approved	F-08, F-19; T-08; P-01

p99 means that 99% of completed measurements meet the stated value; failures and timeouts must also be reported. These are statistical objectives, not hard real-time guarantees.

TX-02 is another output branch of the same sample, not an extra request stream. A 60 Hz display cannot show a separate frame for every 200 Hz sample. Its test selects clearly identifiable events. Samples without an eligible cue have no **t1**. Errors are recorded as errors, not sporting misses or lost points.



Independent outputs after commit. Invalid input -> explicit error or counted local timeout.

Rejected update -> rollback. Unknown commit -> pause, recover; suppress expired controller cue.

Both output branches are independent after commit; display need not wait for controller feedback. If a result is lost, the controller still accounts for its timeout. A commit discovered later may remain in match history, while its expired controller cue stays suppressed.

The Pi processes up to **eight inputs per 5 ms batch**, at most two per controller, with four queued inputs/controller and one state batch in flight. Every valid selected sample retains its required query/response. Periodic empty-query batches keep the ball moving. Only the Pi calculates changes; the service accepts or rejects the **whole** match update. Rejection also restores candidate score, avatar, contact and cue state. [F-02, F-04, F-05, F-09, N-03]

Stage budgets

The following is the complete routine-path allocation. Its finer category split is proposed under P-03. Input means reading data; action includes processing and waiting; transmission/reception include network and scheduling delays; output ends at actual electrical activation. All values are milliseconds.

Stage / responsible role	Input	Action	Transmit	Receive	Output	Total
Read and encode IMU / controller owner	0.20	0.30	0	0	0	0.50
Controller → Pi / network and Pi I/O owners	0	0	3.50	1.50	0	5.00
Wait for batch / Pi scheduler owner	0	5.00	0	0	0	5.00
Select and validate pose / fusion owner	0.25	0.25	0	0	0	0.50
Required query and response / Pi and state-service owners	0	0.35	0.30	0.35	0	1.00
Calculate ordered results / physics owner	0	0.50	0	0	0	0.50
Update and commit response / Pi and state-service owners	0	0.35	0.30	0.35	0	1.00
Feedback → controller / network and controller owners	0	0	3.50	1.50	0	5.00
Validate and activate / controller output owner	0	1.00	0	0	0.50	1.50
TX-01 total	**0.45**	**7.75**	**7.60**	**3.70**	**0.50**	**20.00**

This gives **18.5 ms to response receipt**, **20 ms to electrical output**, and **3 ms batch service**, before variation/retries. A 10 ms planning margin is reserved; the separate electrical objective remains 40 ms. T-06 also proposes a 15 ms p99 diagnostic target from physics decision to electrical output. Measure complete intervals directly: adding individual stage percentiles does not prove their p99.

TX-02 shares the prefix through accepted commit. Its additional budget is:

Stage / responsible role	Categories	Proposed budget
Shared sensor-to-commit prefix / controller, network, Pi and state owners	Input 0.450 + action 6.750 + transmission 4.100 + reception 2.200; conservative full commit envelope	13.500 ms
Local publication / state-service and Unity owners	Transmission 0.400 + reception 0.600	1.000 ms
Frame wait and scene update / Unity owner	Action	18.667 ms
Submission to photons / presentation owner	Output, including scanout/panel delay	20.000 ms
TX-02 total		**53.167 ms**

The P-01 proposed 80 ms screen objective leaves 26.833 ms planning headroom. Publication does not actually wait for the Pi's commit acknowledgement. These display/frame assumptions need measurement on the selected hardware. Full physical movement-to-vibration latency remains open under T-08.

4. Global time

Applies: F-10, F-14, F-15; N-02, N-11; T-03, T-04, T-05, T-07; P-02.

The **Pi monotonic clock** is the common reference. Map each controller/laptop clock with **T_pi = a × local_time + b**: **b** corrects offset and **a** corrects drift. Preserve original timestamps, source/reference sessions and the mapping version. [F-15, N-11]

Property	Required/proposed value
Timestamp representation	Integer microseconds; this does not prove microsecond accuracy
Sampling/simulation granularity	200 Hz per controller / 5 ms physics step
Cross-device accuracy	Proposed justified uncertainty ≤2 ms per source; two independent sources can give ±4 ms relative uncertainty, plus capture errors
Clock maintenance	Proposed refresh every 1 s; expire after 5 s, or sooner if uncertainty is too large
Camera timing	Separate proposed capture-time uncertainty ≤8 ms
Local measurement accuracy	P-02 proposal: timestamp quantum ≤10 µs and combined interval uncertainty ≤0.10 ms; verify marker skew and instrument timebase

Use complete four-timestamp clock exchanges and retain rejected observations. Do not assume equal network delays or calculate one-way delay as RTT/2. A restart invalidates affected future mappings; a Pi restart creates a new reference epoch. Keep earlier valid history. Unknown/expired mappings prevent cross-device freshness claims, while local sample accounting continues.

Measure **tr-t0** and **t1-t0** on the **same ESP32 clock**; they need no inter-device synchronization. Keep camera capture, arrival, inference completion and simulation time separate. Pose must align with historical IMU data. Apply all proposed gates, including uncertainty: IMU age ≤30 ms, pose age ≤80 ms, prediction ≤50 ms and alignment error ≤25 ms. Recheck after network waiting. The debug view must show uncertainty and leave unsupported timing unresolved.

The extra IMU threshold interrupt is only a diagnostic **t_trigger** marker. Explicitly link it to the first regular sample captured at/after it. It does not create a transaction, replace **t0**, change priority or reset expiry. The MPU-6050 has one shared interrupt line, so ambiguous edges remain unresolved. [F-02, F-14]

5. Interfaces and messages

Applies: F-02-F-12, F-15, F-20; N-03, N-07; T-09, T-10.

Use the accepted v0.3 contract, whose exact version string remains **0.3-proposed**. All participants must use the same schema/profile hashes. The complete interface summary required for this design is included below. [N-07]

Link	Transport	Messages / essential content	Failure handling
Controller ↔ Pi	UDP; Pi input 41000; negotiated controller reply endpoint	**controller_input**: transaction/sample/timestamps, IMU, association/calibration. **feedback_result**: matching transaction, outcome, committed event and cue eligibility	Input sent once. Reject wrong/old/duplicate replies; local expiry 50 ms
Tracking → Pi	UDP 41020	**skeleton_frame**: person/frame, capture time, joints, quality. **frame_status**: visible people/readiness/drop count	Missing/stale/invalid pose disables hit eligibility
Pi ↔ state service	Dedicated TCP 41010	**ball_state_request** / **ball_state_response**: per-sample queries and common versioned snapshot	Identical query retry at 2 ms; expiry 5 ms
Pi ↔ state service	Same dedicated TCP	**state_update** / **state_commit_result**: base version, full match, ordered events, acceptance/rejection	Identical retries at 2/4 ms; pause uncertainty at 8 ms; resolve via separate control

Link	Transport	Messages / essential content	Failure handling
State service → Unity	Bounded local TCP subscription	**state_publication**: committed match/version/events	Slow rendering must not block state handling; retain event/loss accounting
Control and clocks	Separate TCP: Pi 41040, game 41041, tracking 41042	**hello**/**hello_result**, **association**, **match_setup**, **control_command**, **clock_exchange**	Explicit setup/installation ACK and resume; reconnect/restart revalidates sessions
Nodes → debug	Planned UDP 41030; debug control TCP 41043	Buffered traces, clock maps, marker links, source/loss counters	Never block gameplay; missing evidence stays visible

UDP envelopes are ≤1,200 bytes. State/publication envelopes are ≤8,192 bytes with a four-byte big-endian length prefix. State framing has an 8 ms assembly deadline, three outbound frame slots and one inbound buffer. Control has a separate 65,536-byte outer limit and 1 s assembly deadline. Invalid/oversized/incomplete frames are rejected or closed before partial state effects; TCP recovery never extends transaction deadlines.

Preserve transaction, sample, frame/person, query, batch, event, session/epoch, association/calibration and state-version identities. Retain query/commit outcomes for 2 s, at most 512 batches; full retention causes explicit overload, not unsafe eviction. Changed bytes under an existing batch key are invalid. Confirm player/controller bindings explicitly; never silently assign a nearby person. Normal doubles requires four confirmed A1/A2/B1/B2 players. [F-11, F-20]

Message rules: wire physical values use SI units and must be finite. Capture times keep their clock domain and validity; unknown time is not zero. Joint coordinates are normalized image coordinates, not metric depth. Setup binds four unique controller/person/slot/avatar identities, playing hand, association/calibration revisions and immutable rules/configuration. A response echoes the complete transaction or query/batch key. Hit/miss feedback identifies an accepted game session, writer epoch, batch, event index, version and tick. Error outcomes are invalid_input, tracking_unavailable, stale_input, state_conflict, timeout, overloaded or internal_error; none may masquerade as hit/miss. [F-03-F-11, F-13, F-20; N-07]

Validation rejects incompatible versions/hashes, unknown or reused sessions, invalid lengths, missing identities and nonfinite values before gameplay effects. Counters/timestamps use integer values bounded to 2^56-1; session IDs are 16 bytes and hashes 32 bytes. A state update holds at most eight decisions and sixteen rule events. The source history retains 256 IMU samples/controller and eight poses/person for bounded historical alignment. [F-09, F-10; N-02, N-03, N-07]

6. Proposed fault hypothesis

Applies: F-06, F-07, F-09-F-15, F-19, F-20; N-02-N-07, N-09, N-11.

No approved standalone fault hypothesis was found. The following fault families define this report's test scope. Test single fault families first, then the stated combinations. Scope and exclusions need team approval.

ID	Fault family	Required reaction
FH-01	Network loss, delay, jitter, duplication, reordering or AP outage	Count failures, reject mismatches/late work, enforce expiry and revalidate reconnect
FH-02	Invalid messages, incompatible versions, oversized/partial TCP frames or stalled stream	Bounded validation/close; no partial state/output; independent recovery control
FH-03	Lost commit result, stale state, conflicting writer or restart around commit	Exact retries, atomic rejection/rollback, pause uncertainty, fence old sessions/epochs
FH-04	Sensor/pose failure, old capture, bad calibration or ambiguous player binding	Explicit invalidity and suspended eligibility; no fabricated miss, point or depth
FH-05	Clock drift/asymmetry, expired/missing map or clock restart	Fail cross-device freshness checks; preserve local metrics, raw history and epochs
FH-06	Queue overflow, CPU/GPU/render load, low battery, thermal or actuator fault	Bounded overload/pause and safe outputs; no hidden sample-rate reduction
FH-07	Collector/trace/marker loss, shared-IRQ ambiguity or observation overhead	Gameplay stays independent; count missing evidence and unresolved links
FH-08	Missing/late publication, wrong team view or missing photons	Display only committed state; count missing presentation and preserve event identity

TCP packet reordering is not application-message reordering. Inject TCP message omission/replay at a complete-message boundary; deleting stream bytes tests framing corruption. Proposed exclusions are malicious/Byzantine peers, automatic failover and arbitrary combinations of permanent hardware failures. The design promises explicit failure/pause, not continued play under every fault.

7. Integration test specification

Applies: F-01-F-20, N-01-N-07, N-09-N-11, T-01-T-10, P-01-P-03. Each IT row inherits the shared measurement and evidence procedure; section 8 gives its exact requirement mapping.

All tests below are planned, not run. Missing hardware, runtime, approved fixture or instrument makes a test blocked.

Shared setup, measurement and reporting

These instructions apply to every test in the table:

1. Record exact code/schema/profile/tool versions, hardware, topology/routes, workload, seed, calibration, player bindings and valid clock maps. Verify safe power/probe connections, bounded queues and setup/ACK/resume. A test instrument's resolution alone does not prove accuracy.

2. Record **t0** at the regular controller boundary, complete reception **tr**, validation completion and actual LED/motor drive **t1**. Capture IRQ/GPIO markers and actual drive with a common logic analyzer; characterize marker skew. Preserve IDs through pose selection, query, commit, feedback and output. An enqueue call is not electrical output.

3. For TX-02, observe keyed display patches with photodiodes on the same acquisition timebase as **t0**. Join each patch to the committed event, render frame and actual scene change in both views. Frame submission is not a photon timestamp. Record scan position and optical threshold.

4. Apply the test stimulus, audit expected behavior and save raw logs, clock maps, electrical/optical records, injected identities/parameters and verdict. Use role labels, device/session IDs and anonymous player slots only; exclude credentials, student names and identifiable participant images from submitted artifacts. Remove faults/load, resolve pending commits, restore safe outputs and valid sessions/readiness; then run a 60 s clean recovery check.

Baseline: **three 10-minute runs at each controller count 1-4, for A and B**; 24 runs total, with separate declared warmup. Keep ordinary 200 Hz traffic and continuous tracking. One-to-three-controller tests use explicit diagnostic fixtures, not an assumed doubles match or silent AI players. Fault tests repeat at one and four controllers in both variants; use seed **20261110 + test number**, at least 100 affected identities per subcase unless a duration is stated. Real measurements and synthetic fixture checks are labeled separately.

Report counts, median, p95, p99, observed maximum, deadline misses, technical failures and missing evidence. Define nearest-rank percentiles on completed intervals. Count every started transaction once as gameplay result, explicit failure, local failure/timeout or pending. A timeout is an unknown completion time beyond the cutoff, not a fabricated 50 ms response. Report all-response and successful-only populations separately, and electrical timing only for eligible cues; missing eligible output still counts as failure.

Clean TX-01 gates: response p99 ≤30 ms, eligible electrical p99 ≤40 ms, ≤1% late-or-failed transactions, zero wrong bindings/duplicate commits/new expired cues and zero lost acceptance logs. Count each failing transaction once even if several criteria fail. Measure batch-service p99 ≤4 ms separately. Fault tests instead check bounded failure/recovery. TX-02's proposed sparse-event gate is p99 ≤80 ms per view with zero missing selected events; a general gameplay display policy remains open.

Test cases

Each row inherits the setup, hardware measurement, correlation, evidence and cleanup above. The last column gives that test's additional pass/fail checks. Roles are responsibilities, not named team members.

Test / transaction / owner	Faults and reproducible procedure	Additional pass/fail checks
IT-01: full path · TX-01 · integration owner	No injected fault. Run the baseline matrix. Use approved expected-state fixtures and known hit, completed-miss and noncue motions; trace every valid sample through query, accepted commit and response	Clean gates met; correct fixture outcomes; no missing logical query, duplicate cue or invented **t1**. Preserve electrical LED/motor timestamps separately
IT-02: display · TX-02 · presentation owner	Nominal FH-08 checks. Select 200 identifiable committed events per active controller, ≥200 ms apart globally. Capture both patches and matching scene frames while 200 Hz traffic continues	Both views show the correct committed event/state; complete optical evidence and proposed screen gate. Missing photons cannot be replaced by submission times

Test / transaction / owner	Faults and reproducible procedure	Additional pass/fail checks
IT-03: network · both · network owner	FH-01. Separate 60 s conditions: added delay 5/15/30 ms; uniform jitter 0-20 ms; loss 1/5/20%, independently per direction. Replay/reorder complete messages by 10 ms; disconnect AP for 2 s	Every source transaction accounted; no duplicate/mismatched/expired cue. Recovery revalidates endpoints. Preserve a committed event even if its feedback is lost
IT-04: state/restart · both · state-service owner	FH-03. Separately omit input, query, snapshot, update, commit result and feedback; include first-copy and all-retries cases. Delay known acceptance beyond **t0+60 ms**; replay stale/changed-key updates. Restart Pi/state/controller before update and after commit/before reply	Query 2/5 ms and commit 2/4/8 ms policies hold; one atomic commit, full candidate rollback on rejection, no next batch while uncertain. Old epochs fenced; UNKNOWN_BATCH requires recovery/reset; no late controller cue
IT-05: input/identity · both · tracking/fusion owner	FH-04. Occlude wrist/freeze/unplug camera for 2 s; inject saturation and failed sensor reads. Test uncertainty-inclusive age/alignment gates 1 ms below/above each limit. Move calibration, swap/reboot controller, reuse tracker ID and deliver an old association ACK/feedback	Invalid data yields explicit technical error, not miss/point. Historical IMU alignment retained; all freshness gates apply after waits. Unique binding, new-session confirmation, matching install/setup ACK and resume required
IT-06: clocks · both · time owner	FH-05. Labeled fixtures: offsets ±100 ms, drift ±100 ppm and directional exchange delays (0,10)/(10,0) ms. Block refresh for 6 s; exceed 2 ms uncertainty earlier; restart a source then Pi. Compare real clock bounds with independent reference where available	Error covered by declared bounds; invalid future maps rejected, valid history preserved. Local **tr-t0**/**t1-t0** remain usable. Viewer shows overlapping ±2 ms intervals as uncertain order; no RTT/2 assumption
IT-07: load/safety · both · integration/hardware owner	FH-06. Exercise batches of 0/1/4/8, third input/controller and fifth queued input. Add a labeled 10 s 1,000-input/s/controller synthetic burst; separately stall Unity 100 ms every 2 s for 60 s. Run 120 s Pi/Windows CPU/GPU load toward 90% observed utilization; record temperatures/throttling. Test reviewed low-battery/driver controls and overlong pulse requests	Queue/batch/retention limits hold; no second physics writer or hidden rate reduction. Catch-up ≤4 steps, pause above 20 ms lag. No late/replayed onset; pulse cap ≤50 ms, safe default-off; unavailable part/threshold choices block hardware subcases
IT-08: diagnostics/markers · both · debug/time owner	FH-07. Stop collector 10 s, overflow one bounded trace ring, remove map/link records and trigger near sample boundaries/shared-INT overlap. Compare three 120 s pairs with full versus minimal logging/markers, using common external capture	Gameplay independent; loss/unresolved evidence explicit. Trigger links use actual sample identity, never nearest arriving reply; no extra transaction or changed priority. Quantify overhead and instrument uncertainty; incomplete logs cannot pass acceptance
IT-09: doubles/views · both · gameplay integration owner	FH-04/FH-08 plus nominal rules. Four confirmed humans: lift/overhead serve; cancelled toss/attempted miss; first/second fault/let; line/net/out/second bounce; deuce/tie-break/end changes; simultaneous swings and technical pause. Use versioned expected-rule fixtures. Separate negative view fixtures inject an old viewport version and swapped team-camera transform	Pi alone calculates movement/score; atomic matching state in both normal views; stable A1/A2/B1/B2 through end changes. Negative view mismatches must receive a failed-view audit verdict. No automatic hit, silent reassignment or technical-failure point. Periodic query/commits move ball between human actions
IT-10: protocol/bounds · both · interface owner	FH-02/FH-03/FH-06/FH-08. Send malformed/nonfinite/wrong-version/hash/identity fixtures, 1,201-byte UDP and 8,193-byte state frames; split/coalesce valid TCP frames and hold incomplete state 20 ms/control 1.2 s. Stop subscriber reads 2 s. In a labeled controlled-time host fixture fill 512 live retained batches and query after 2 s expiry	Bounded reject/close, no partial commit, no writer blocked by rendering. State 8 ms/control 1 s assembly limits remain distinct; exact retries deduplicate, changed payload rejects and expired status is UNKNOWN_BATCH. Host fixtures do not prove hardware throughput

Also run three named combinations after their single-fault tests: **lost commit result + stopped collector**, **15 ms feedback delay + high Pi CPU load**, and **frozen pose + expired clock map**. Preserve each constituent reaction. Other combinations remain unverified. Before execution, each test fixture records its explicit input, expected result and configuration; an unavailable oracle blocks that subcase. [N-05, N-07]

8. Traceability

Each row below defines an existing project requirement and maps it to a transaction, node and test. P-01-P-03 are new review proposals, not accepted baseline changes. All timing limits remain proposed/unmeasured. C = controller, P = Pi, A = AP/network, T = tracking, S = state service, U = Unity, D = debug. Both = TX-01 and the selected TX-02 branch. Review-only requirements do not create a sensor transaction.

Req. ID	Definition	Slice	Node	Test / evidence
F-01	Support 1-4 wireless battery controllers.	Both	C, P, A	IT-01, IT-03, IT-07
F-02	Keep one transaction per regular IMU sampling boundary.	Both	C, P	IT-01, IT-07, IT-08
F-03	Fuse active camera body/joint observations with IMU on Pi.	Both	C, T, P	IT-05, IT-06, IT-09
F-04	Give every valid selected input a matching ball-state query/response.	Both	P, S	IT-01, IT-04, IT-10
F-05	Pi calculates hit/miss and resulting ball motion.	Both	P, S	IT-01, IT-04, IT-09
F-06	Account for every generated transaction, including local failure.	Both	All	IT-01 to IT-10
F-07	Distinguish hit, completed-miss and fault feedback.	TX-01	C, P, S	IT-01, IT-04, IT-07
F-08	Store one committed match on Windows; present it in separate Unity.	Both	P, S, U	IT-02, IT-07, IT-10
F-09	Use one ordered physics writer for concurrent and periodic work.	Both	P, S	IT-04, IT-07, IT-09
F-10	Preserve original capture timestamps and source identities.	Both	All	IT-01, IT-05, IT-06, IT-08
F-11	Handle connection, association, calibration, loss and restart explicitly.	Both	C, T, P, S	IT-03, IT-04, IT-05, IT-06
F-12	Provide cross-system debugging, tracing and fault-test support.	Both	All, D	IT-08; Mx review
F-13	Treat webcam gestures as calibrated 2D/2.5D, not measured 3D.	Both	T, P, U	IT-05, IT-09
F-14	Keep the hardware threshold marker diagnostic; preserve sampling.	Both	C, D	IT-08
F-15	Map clocks to Pi with offset/drift correction and visible uncertainty.	Both	All, P, D	IT-06, IT-08
F-16	Play four-human 2v2 doubles with automatic avatar movement.	Both	C, T, P, S, U	IT-09
F-17	Recognize assigned-server lift/toss and overhead serve gestures.	Both	C, T, P, S, U	IT-09
F-18	Apply normal tennis scoring; technical failures do not award points.	Both	P, S, U	IT-04, IT-05, IT-09
F-19	Show one committed match in two elevated rear team views.	TX-02	S, U, display	IT-02, IT-07, IT-09
F-20	Confirm unique controller/person/team-slot assignment before resume.	Both	C, T, P, S, U	IT-05, IT-09
N-01	Meet controller-local response/electrical timing objectives.	TX-01	C, P, S	IT-01, IT-03, IT-04, IT-07
N-02	Check sensor age, historical alignment and timing uncertainty.	Both	C, T, P	IT-05, IT-06
N-03	Bound memory, queues, state batches and catch-up work.	Both	C, P, S, U, D	IT-07, IT-08, IT-10
N-04	Report complete populations, failures, timeouts and observation loss.	Both	All	All tests

Req. ID	Definition	Slice	Node	Test / evidence
N-05	Make builds and test runs reproducible with exact version manifests.	Both	All	All test manifests
N-06	Keep logging nonblocking; measure its overhead and losses.	Both	All, D	IT-08
N-07	Use compatible, versioned interfaces and exact contract hashes.	Both	All	IT-04, IT-05, IT-10
N-08	Write project content and diagram labels in English.	Review	Report	Document review
N-09	Use safe battery/driver design, default-off and bounded pulses.	TX-01	C	IT-05, IT-07
N-10	Separate statistical objectives from hard real-time guarantees.	Both	All	Timing/results review
N-11	Qualify cross-device durations using valid bounded clock maps.	Both	All, P, D	IT-06, IT-08
T-01	Response p99 <=30 ms from t0; local expiry 50 ms.	TX-01	C, P, S	IT-01, IT-03, IT-04
T-02	Eligible electrical onset p99 <=40 ms; no new onset at/after expiry.	TX-01	C, P, S	IT-01, IT-04, IT-07
T-03	Sample at 200 Hz/controller: nominal 5 ms, 800 inputs/s for four.	Both	C	IT-01, IT-07, IT-08
T-04	5 ms physics step; service p99 <=4 ms; <=4 catch-up steps; pause >20 ms lag.	Both	P, S	IT-01, IT-07
T-05	Measure t0-to-Pi input delivery; routine allocation 5.5 ms.	Both	C, A, P	IT-01, IT-03, IT-06
T-06	Measure decision-to-electrical onset; proposed diagnostic p99 <=15 ms.	TX-01	P, S, C	IT-01, IT-04, IT-06
T-07	IMU age <=30 ms; pose <=80 ms; prediction <=50 ms; alignment <=25 ms.	Both	C, T, P	IT-05, IT-06
T-08	Measure actual physical/optical onset separately; full movement/vibration limit open.	TX-02 / optional	Sensors, display, actuator	IT-02, IT-08; CR-03
T-09	Query: identical retry at 2 ms; expire at 5 ms.	Both	P, S	IT-04, IT-10
T-10	Commit: identical retries at 2/4 ms; pause ambiguity at 8 ms.	Both	P, S	IT-04, IT-10
P-01	New proposal: selected visible event p99 <=80 ms/view, zero missing selected events.	TX-02	S, U, display	IT-02; CR-03 decision
P-02	New proposal: native quantum <=10 us, measured interval uncertainty <=0.10 ms.	Both	C, instruments	IT-02, IT-06, IT-08; CR-03
P-03	New proposal: fine budget category allocations in section 3.	Both	C, P, S, U	IT-01, IT-02; CR-03

9. Consistency review and acceptance

The author checked ownership, mandatory messages, budgets, clocks, test coverage and the submission rules. Fixed means corrected in this report. Accepted requires a recorded team risk decision; no such decision was supplied. [N-04, N-05, N-07, N-10]

Finding	Requirement	Status	Owner role	Correction or closure
CR-01: missing submission format, title and team	N-05, N-08	fixed	Documentation	One PDF, prescribed team filename, team/milestone/date title, role-only metadata
CR-02: repository dependencies and undefined requirement IDs	F-10, N-05, N-07	fixed	Documentation	Required tables, message extracts, diagrams and all requirement definitions are included; no external file lookup is needed
CR-03: new display, accuracy and budget proposals	T-08, N-01, N-02; P-01-P-03	deferred	Time / UI	Team must approve or revise 80 ms display target, selected-event policy, measurement accuracy and fine budget allocations; full physical movement/vibration target stays open
CR-04: fault scope and execution prerequisites	F-06, F-12; N-03, N-05, N-09	deferred	Integration	Approve hypothesis/exclusions, expected-result fixtures, actual hardware/driver limits, diagnostic slot setup and bounded trace/publication policies
CR-05: sample/marker and ownership ambiguity	F-02, F-04, F-08, F-09, F-14	fixed	Physics / control	One regular sample remains one transaction; queries remain mandatory; Pi calculates, service stores and Unity presents
CR-06: percentile/clock claims could imply hard bounds	N-04, N-10, N-11; T-01-T-07	fixed	Time owner	Complete populations, direct durations, conservative mapping bounds and censored timeouts are explicit
CR-07: exact-byte deployment checks differ on Windows	N-05, N-07	deferred	Contract tooling	Align distributed contract bytes and portable freshness checks before claiming target compatibility; do not silently change accepted identities
CR-08: approvals and separate Mx missing	F-12; N-05	deferred	Review + debug	Record role-only design decision and provide acceptance of the separately versioned Mx concept

The PDF is an anonymous design/test specification, not a claim that hardware tests pass. Exact tool/device manifests are created when those tests become executable. The team remains responsible for the technical content, including AI-assisted drafting. [N-04, N-05, N-10]

Milestone decision	Present evidence	Status / required action
Gate 2 consistency review	Author review and findings above; no independent team decision supplied	Review complete by author; team design approval pending. Resolve or explicitly accept deferred design risks and record roles, revision, decision and date; no student names
Mx - Integration Test Debugging Concept	Required enablement is defined below; separate accepted artifact not supplied	Pending artifact/acceptance
M3	Requires acceptance of this report AND Mx	Not reached; neither acceptance is recorded

Mx must define bench/startup procedures, exact timestamp/probe boundaries and calibration, ID joins, Pi clock-map evidence, packet/application fault hooks, bounded nonblocking logging and loss counters, and the reproducible run bundle/auditor. Sections 6-7 state the requirements consumed by this report; they do not substitute for Mx acceptance. [F-10, F-12, F-14, F-15; N-04-N-07, N-11]